few entries, and sum up the values in each list by entry.

For instance, if Daisy Duck is assigned the $1, 50¢, 25¢ and 1¢ coins, then all the machines will send her a short list that looks like this:

```
25¢: 1.034 pieces
$1: 11823 pieces
50¢: 53982 pieces
1¢: 321 pieces
```

Figure 13.5 shows the shift in paradigm: while beforehand whoever had to make sense of all the lists was the bottleneck of the computation, now, introducing a new intermediate level in the workflow, and breaking down the work so that each entity at this level has a limited amount of work to do, make the whole difference.

The key, though, is that the results outputted at level 1 can be partitioned in groups and each of these groups can then be handled separately at level 2.

13.2.2 First Map, Then Reduce

Time to abandon our comics' heroes for a more life-like example, where both levels of this parallel computation would be performed by machines. The operation at level 1 is called *Map*, because it *maps* each entry in the input dataset (more exactly, in the portion of the dataset handled by a machine) into something else, extracting the information that's relevant for computing the final result; the counting